Metric Holonomy, Statistical Selection and Finite-Carrier Obstructions

An externally sourced, bounded mathematical assessment

Research direction: **Kirtus Justus**, Port Coquitlam, British Columbia, Canada

AI-assisted mathematical assessment: **ChatGPT (GPT-6), Director G**

12 November 2026, B.C. Pacific Time

Gate status. Initial External Grounding, **G1 OPEN**. No physical ontology, quantum emergence, classical emergence, or complete correspondence map is established. PCP Version 1.0/ACR was read and applied as an *absorbed*, non-host-enforced review methodology. All mathematical constructions in this note are conditional on explicitly imported geometry.

Source boundary. The accessible baseline, PCP Enterprise packet and toy counterexample packet passed their bundled local structural verifiers. The separate Director Z Z1 baseline and manuscript G7 were described in a transmitted disposition but not attached here; their proofs and numerics are *not* certified by this assessment. No sub-agents or independent reviewers were actually deployed. More precise provenance, external-source limits and ACR logging are in the accompanying archive.

1 An exact metric-only phase nonidentification

Let $\langle\cdot,\cdot\rangle$ be the standard Hermitian inner product, conjugate-linear in its first variable. For normalized nonorthogonal rays set

$$P(u,v) = |\langle u, v\rangle|^2, \qquad B(x,y,z) = \langle x,y\rangle\langle y,z\rangle\langle z,x\rangle. \tag{1}$$

A full Fubini–Study distance function is $d([u],[v]) = \arccos\sqrt{P(u,v)}$.

Theorem 1 (Signed phase cannot be reconstructed from probabilities alone)**.** *No function invariant under all transition-probability isometries of complex projective space can universally recover the oriented Bargmann phase $B/|B|$.*

Proof. In $\mathbb{C}^2$ take

$$x = \begin{pmatrix}1\\0\end{pmatrix}, \quad y = \begin{pmatrix}\sqrt{3}/2\\1/2\end{pmatrix}, \quad z = \begin{pmatrix}\sqrt{3}/2\\ i/2\end{pmatrix}.$$

All three vectors have norm one. Direct multiplication gives

$$(P(x,y), P(y,z), P(z,x)) = \left(\frac{3}{4}, \frac{5}{8}, \frac{3}{4}\right), \quad B(x,y,z) = \frac{9+3i}{16}.$$

Conjugate *all* three vectors. Complex conjugation is an isometry of the complete projective transition-probability metric: every $P(u,v)$ remains unchanged. But the ordered product changes to $(9-3i)/16$, which has the opposite nonzero phase. Any metric-only invariant would have to take the same value on both configurations, a contradiction. The example embeds into $\mathbb{CP}^n$ for every $n \geq 1$. □

The underlying ambiguity is also described by Li, Wagner and Zhang (2026) [2]; this construction claims no discovery priority.

Remark 1 (The extra datum hidden in complex determinants)**.** The Levi-Civita connection is determined by the *real* Riemannian metric. A complex determinant $\det_{\mathbb{C}}\mathrm{Hol}$ additionally requires a selected parallel complex structure J. Replacing it by $-J$ conjugates that scalar. The resulting real orientation changes by $(-1)^n$ in complex dimension n, so *even-dimensional* $\mathbb{CP}^n$ does not resolve the ambiguity by specifying real orientation alone. A signed Kähler form or equivalent operational warrant must be supplied and justified, rather than inferred silently from transition probabilities.

2 A positive conditional theorem: determinant holonomy

The following is a statement *inside* conventional complex quantum geometry, not a construction of that geometry.

Proposition 1 (Determinant/Bargmann correspondence)**.** *Assume $\mathbb{CP}^n$ has its standard Fubini–Study Kähler metric and its* chosen *complex structure J. For a projective geodesic polygon γ whose successive representative rays are nonorthogonal, write $B = \prod_{j=0}^{m-1} \langle \psi_j, \psi_{j+1} \rangle$ with $\psi_m = \psi_0$. With the corresponding orientation and Pancharatnam conventions,*

$$\boxed{\det_{\mathbb{C}} \mathrm{Hol}_{T^{1,0}\mathbb{CP}^n}(\gamma) = \left(\frac{B}{|B|}\right)^{n+1}.} \tag{2}$$

Proof. In the usual affine coordinate $z \in \mathbb{C}^n$, put $q = 1 + \|z\|^2$. The Fubini–Study metric is

$$g_{j\bar{k}} = \frac{q\delta_{jk} - \bar{z}_j z_k}{q^2}, \qquad \det g = q^{-(n+1)}.$$

The tautological line $L = \mathcal{O}(-1)$ has local squared Hermitian norm q; its dual $L^* = \mathcal{O}(1)$ has squared norm q^{-1}. The projective Euler sequence identifies the determinant holomorphic line of $T^{1,0}\mathbb{CP}^n$ with $(L^*)^{\otimes(n+1)}$. The displayed determinant shows that the induced Chern connections agree in these local frames; any constant normalization factor is connection-invisible. On a Kähler manifold the tangent Chern and Levi-Civita connections agree. Thus the tangent complex determinant holonomy equals the $(n+1)$-th power of the dual-tautological holonomy.

For the chosen ordered sequence, geodesic Pancharatnam transport keeps adjacent representatives in phase. On closing the polygon the *tautological* line obtains holonomy $(B/|B|)^{-1}$, so the dual line obtains $B/|B|$. Equation (2) follows. Reversing orientation conjugates both sides. □

An independently coded spherical transport check for $n = 1$ compared actual great-circle Levi-Civita rotations to $2 \arg B$ for **96 seeded triangles**, retaining every result. The maximum wrapped angular discrepancy was 1.53143×10^{-15} radians. This is numerical corroboration *only for conventional input geometry*; the code is **not Schild's ladder**, and Director Z's reported $n = 2, 3$ ladder calculations were not available for comparison. Published Schild's-ladder analyses specify convergence/scaling hypotheses [4].

3 The missing statistical theory: classical-spin mirror

The phase space of a classical spin with fixed magnitude is S^2, with round metric, $SO(3)$ symmetry and the nondegenerate signed area two-form. Its Levi-Civita holonomy is $SO(2)$ and its real tangent commutant is

$$\mathrm{Comm}_{\mathbb{R}}(SO(2)) = \{aI + bJ : a, b \in \mathbb{R}\} \cong \mathbb{C}.$$

Rescale its round metric to match the pure-state Fubini–Study sphere $\mathbb{CP}^1$. These *pure-state* geometric invariants coincide. Yet its classical statistical state space can be all Borel probability measures $\mathcal{P}(S^2)$, with measurable effects $0 \le f(n) \le 1$, and probability $\int f \, d\mu$. Disjoint classical filters give additive probabilities, hence no genuine higher-order Sorkin interference.

Define $\mu_{\mathrm{pair}} = \frac{1}{2}(\delta_{\hat{z}} + \delta_{-\hat{z}})$ and let μ_{unif} be uniform area measure. Both have zero first moment and hence map to the *same* qubit density matrix $I/2$ under the conventional Bloch barycentre map. Their classical predictions differ for $f(n) = n_z^2$:

$$\int f \, d\mu_{\mathrm{pair}} = 1, \qquad \int f \, d\mu_{\mathrm{unif}} = \tfrac{1}{3}.$$

Therefore pure-state holonomy, a symplectic form and no third-order interference *alone* do not uniquely fix the statistical theory.

Scope caveat. $\mathcal{P}(S^2)$ has infinite affine dimension; the operational distinguishability metric induced by its *full* classical effect set is not the qubit Fubini–Study metric. This defeats an unrestricted uniqueness assertion but does not refute a narrower finite-dimensional, irreducible Jordan/operational-geometry classification. Such reconstructions require substantial further assumptions [3].

4 A curvature control for real qutrits and complex qubits

Equip both spaces with the same rule $d([u],[v]) = \arccos|\langle u, v\rangle|$. Then

Pure space	Gauss curvature	π_1	full holonomy
$\mathbb{RP}^2$ (real three-level)	1	$\mathbb{Z}_2$	$O(2)$
$\mathbb{CP}^1$ (complex two-level)	4	0	$SO(2)$

Indeed $\mathbb{RP}^2$ is locally the unit sphere whereas $\mathbb{CP}^1$ is the round sphere of radius $1/2$ for this normalization. Their restricted holonomy groups are both $SO(2)$, so their restricted real holonomy commutants are both $\mathbb{C}$. However, intrinsic *local curvature* already separates them, as do full holonomy and topology. Monodromy is thus not their *only* geometric discriminator if the full transition-probability metric has been supplied. This distinction does not derive the metric from anything prephysical.

5 Finite carriers: no-go theorem and area certificates

Theorem 2 (Application of Gelander's limit theorem)**.** *A sequence of finite* homogeneous *metric spaces cannot converge in the Gromov–Hausdorff sense to $\mathbb{CP}^n$ with its standard manifold topology for any $n \geq 1$.*

Proof. Gelander proved that if a manifold is a Gromov–Hausdorff limit of finite homogeneous metric spaces, then it is a torus [1]. For $n \geq 1$, $\mathbb{CP}^n$ is simply connected and is not a torus. Hence the claimed convergence is impossible. This is an *application* of a published result, not a new proof of its hypotheses. □

This excludes exact finite-stage transitive-isometry symmetry *under that convergence notion.* It does not exclude nonhomogeneous discrete carriers, approximate symmetry, or differently defined emergence maps.

Proposition 2 (Finite single-orbit empty-cap certificates)**.** *Let S be a* single finite symmetry orbit *on the unit Bloch sphere, with geodesic covering radius r. If S is a finite $SO(3)$-orbit, then*

$$r \geq \arccos(29/30) \simeq 14.83511^\circ.$$

If S is a finite $O(3)$-orbit (permitting antiunitary-induced isometries), then

$$r \geq \arccos(59/60) \simeq 10.47531^\circ.$$

Proof. If $N = |S|$, covering by N spherical caps of radius r gives $N2\pi(1-\cos r) \geq 4\pi$, hence $r \geq \arccos(1-2/N)$. Finite rotation subgroups are cyclic, dihedral, tetrahedral, octahedral or icosahedral; the three polyhedral orders are at most 60, so their orbit sizes are at most 60 [5]. Cyclic and dihedral orbits, aside from exceptional small groups, lie on at most two latitudes around the unique invariant principal axis. Their north/south-pole and equatorial gaps force $r \geq 45^\circ$. Small exceptions satisfy the displayed bound directly by cardinality.

For a finite $G \leq O(3)$ put $H = G \cap SO(3)$ (normal, index at most two). If H is polyhedral, then $|G| \leq 120$. If H is cyclic or dihedral with unique principal axis, normality forces G to preserve its unoriented axis; an orbit then lies in at most two latitudes and $r \geq 45^\circ$. Exceptional subgroups without that unique axis, notably the order-four dihedral group, have $|G| \leq 8$ and satisfy the weaker displayed estimate by cap area. The two conclusions follow. □

The submitted Director Z summary quotes an uncovered 10.48° cap. The exact $O(3)$ bound rounds to 10.48° but *does not assert* the strict lower bound $r \geq 10.48^\circ$. The source G7 manuscript and its exact assumptions remain uninspected.

6 Implications for the prephysical-to-physical program

None of these constructions identifies a total ontology. A bridge would still need to ground, without unannounced imports: (1) a carrier/limit and state-equivalence rule; (2) operational transition experiments, rather than a stipulated metric; (3) selection of signed phase geometry J versus $-J$; (4) convex states, effects, composition, Born probabilities and instruments; (5) physically calibrated clocks, dynamics and units; and (6) a controlled quantum-to-classical limit with observables and error bounds. The complete, typed directed obligation map, countermodel caveats, source ledger, validator and retained raw trials accompany this paper.

Final disposition: the determinant law is *conditional supported mathematics*. The metric-only phase reconstruction and unrestricted uniqueness proposal have scoped counterexamples. The exact finite-homogeneous Gromov–Hausdorff route faces a published obstruction. No physics has emerged from prephysical premises. The root review remains G1 OPEN, and Director G has not modified or promoted the baseline.

References

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[5] P. Etingof, *Groups Around Us*, MIT lecture notes, chapter on finite rotation groups. `https://math.mit.edu/~etingof/groups.pdf`.